

Adobe's release of Flex 3 in February 2008 was tightly integrated with the release of Adobe AIR. Formerly known as the Adobe Integrated Runtime (and before that by its public code name, Apollo), AIR is Adobe's strategy for offering a universal runtime client that can run local applications on a variety of personal computer systems and other computing devices. With AIR 1.0, Adobe delivered the capability to deploy applications on Windows, Mac OS X, and Linux client systems. With the release of Flash CS5 Professional, Adobe is extending support for AIR to all major smart phones. The roadmap for AIR includes future versions for other cell phones and mobile devices, which eventually would allow AIR desktop applications to be deployed on a more truly universal basis.

AIR applications can be built from many different kinds of assets, but each application's core asset is made up of either Flash-based content, built in either Flash Professional (starting with Flash CS3) or Flex, or HTML-based content. Regardless of which kind of asset is used as the application's core element, any AIR application can use and present HTML, Flash, Flex, or Acrobat PDF content.

The Adobe AIR 2.0 SDK, which includes many new features for integration with local clients, can be downloaded from the Adobe Web site at: www.adobe.com/go/air. In this chapter, we describe the basics of creating and deploying a Flex based desktop application with Flash Builder 4.5 and Adobe AIR.

AIR Architecture

Adobe AIR is installed as a runtime library on your client system. Its purpose is to provide core runtime functionality that's needed by all AIR-based desktop applications, regardless of whether they're built in Flash, Flex, or HTML. Adobe AIR includes a copy of both the Flash Player and a Web browser. AIR 2.0 includes Flash Player 10.1, while the Web browser is an implementation of WebKit 4.0, an open-source Web browser engine. A desktop application deployed on AIR is delivered as an installable archive file with a file extension of .air. After installation, it runs as a local application that's native to the operating system, rather than as a Web-based application. As a result, desktop applications deployed on AIR aren't subject to the same security sandbox restrictions as a Web-based application that's downloaded and run on request from within a Web browser.